Helper card — engineer lane, acute #4 (2026-10-22)

You are running the computer side so the surgeon can stay sterile. Every step is: run the lines, look for ONE pass line, say "PASS" or read the FAIL line aloud. Terminal = PowerShell in `C:\Users\brets\Documents\Repositories\TDTClosedLoop`. First thing in every new terminal:

```
Set-ExecutionPolicy -Scope Process Bypass
$py = "C:\Users\brets\Documents\Repositories\PythonIntanAnalysis\.venv\Scripts\python.exe"
.\rig\0_preflight.ps1 -InputChannels 64        # prints PASS: droppedControlTicks=0
```

Two terminals are used for every stim run: **A** = controller server, **B** = loop. Order is always: start A, start B and wait for the `go` prompt, **then** start the Synapse recording, **then** type `go` and turn the Synapse stim-enable ON right away. When B prints `Summary:` the run is over: stop the Synapse recording, do the flash-drive round-trip (copy the new block folder from the Synapse tank onto the drive, carry it over, copy it into `C:\Users\brets\Desktop\Data\Acute_102226\`; 2-3 min), then run the check lines; they find the newest block by themselves and print its name on the first line — check it is the recording you just stopped. **Only the saline block (row 0b), the gate blocks (rows 1-2) and the quiet block (row 3) are carried over right away; every other block is batched to the WRAP. Nothing else runs on this PC while a run is going** (no analysis, no browser).

Every row's exact lines come from the runcard: `& $py rig\rig_runcard.py --phase <phase0|cgate|gate|quiet|thwack|burst|gap|dutyid|fit|refs|arm|rsweep|drift> --day 2026-10-22 [--short] [--resp 1,3,4,6 --ctrl 34,40 --baseline <V> --blacklist 33,36 --block <dir>]` prints terminal-A / terminal-B / check lines with every blank filled in (and writes them to `day_2026-10-22\runcard_<phase>.txt`). After every `Summary:` the first check line is `run_audit.py` → **one line starting** `AUDIT <run>: PASS` (read `WARN`/`FAIL` lines aloud).

#	When the surgeon says	Run	PASS line to look for
0	"set up" (before induction)	one window: `.\rig\physio_timer.ps1 -Day 2026-10-22 -EveryMin 15` (every beep: read heart rate + body temperature off the PhysioSuite, type e.g. `HR 310 T 37.2`); the Phase-0 lines of `rig_runcard --phase phase0`; flash drive empty and at the Synapse PC	`net_diag` PASS; `find_rz2` one device; selftest `8/8`; preflight PASS
0b	**"array check"** (both arrays in saline, 60 s block recorded and carried over)	`& $py rig\array_health.py --newest --context saline --n-channels 64 --label "cortical saline bench"`	`ARRAY HEALTH: OK` — read the whole line (own-signal N/64, pair-corr, dead banks). `BROKEN` = say so at once: the array must be reseated or swapped BEFORE induction
0c	**"wire gate"** (thalamic array in saline)	`rig_runcard --phase gate --short --depth 0` → A, B → after Summary: `run_audit`, `check_impulse_delivery … --design-meta design_rungate_short_meta.json`	`AUDIT gate0: PASS`; `DELIVERY VERIFIED`, 0 missed; **and the IZ2 front-panel LEDs were GREEN, never RED, during the run** (say which)
1	"cortical gate" (after 40 thwacks, recording stopped)	`rig_runcard --phase cgate` → `array_health … --context tissue`, `placement_gate --mode cortical --newest`, `ingest_block`	`ARRAY HEALTH: OK`; `CORTICAL GATE: GO`; then read THREE lines aloud: `array: …`, `common-mode touch wave … uV`, `re-referenced best chNN … <- control-channel candidate`. If not GO read the `ADJUST:` line aloud (it says whether to move the ARRAY or the REFERENCE)
2	"thalamic gate, position d"	`rig_runcard --phase gate --short --depth <d>` → A, B (`-Run gate<d> -Design design_rungate_short.csv`) → after Summary: `run_audit`, `placement_gate --mode thalamic --newest --min-amp 13`, `check_impulse_delivery … --amps 18 25 30 --design-meta design_rungate_short_meta.json`, `check_dead_time … --y-channel <best ch>`. **At the chosen position once more WITHOUT `--short`** (full 4.4-min design)	`AUDIT gate<d>: PASS`; `THALAMIC GATE: GO` + the per-pair table (read "responsive N, eff rank R, best pair P, best ch C"); `DELIVERY VERIFIED`; IZ2 LEDs green; dead-time `recommended inputDelayTicks = N`
3	"quiet" (60 s recording done)	`& $py rig\ingest_block.py --label quiet --day 2026-10-22`	blacklist list + baseline volts — write both on the whiteboard

#	When the surgeon says	Run	PASS line to look for
4	"thwack " (each battery block)	`rig_runcard --phase thwack --site <SITE> --block <blk>` → its two lines	gallery line with peak µV / best ch / split-half ≥ 0.8
5	"burst probe"	`rig_runcard --phase burst --blacklist <list> --baseline <V>` → A, B → `run_audit`, `check_impulse_delivery … --design-meta design_runburst1_meta.json`, `fit_burst_plant`, `placement_gate`	`AUDIT burst1: PASS`; `DELIVERY VERIFIED`; efficacy line `B=1 1.00 B=3 … B=10 …` + eff-rank per B; gate table with knees
6	"gap scan, pair P"	`rig_runcard --phase gap --gap-pair P --baseline <V>` (prints `DESIGN OK` first) → A, B → `run_audit`, `fit_burst_plant`	`DESIGN OK`; `AUDIT gap1: PASS`; `OPERATING GAP (first ratio >= 0.80): N ticks` — write N on the whiteboard
7	"duty ID, pairs X,Y,Z, gap N"	`rig_runcard --phase dutyid --resp X,Y,Z --gap-ticks N --baseline <V>` (writes + validates the design) → A, B → `run_audit` → `rig_runcard --phase fit --resp X,Y,Z --ctrl <c1>,<c2> --delay-ticks <onset-1>` → its `matlab -batch` line, `check_dead_time`, `check_mimo_rank`, the cpp observer line	`DESIGN OK`; `AUDIT dutyid: PASS`; `VERDICT: OK: centred`; `check_mimo_rank` PASS; `Observer: ACTIVE`
8	"build refs"	`rig_runcard --phase refs --resp … --ctrl … --baseline <V>` → its `build_interleaved_run.py` line, then its `closed_loop_sim.py` line	`wrote 3 run(s)`; `Command varies`
9	"arm MPC rN" / "arm Choi rN" (add `--suffix _late` for 6g, `--suffix _umin --umin <knee>` for 6c)	`rig_runcard --phase arm --arm mpc|choi --run N …` → A line, B line (the y-liveness window opens itself at `go` and BEEPS on DEAD), then `run_audit`, `tracking_metrics`, `ingest_block`	liveness window says `LIVE`; `AUDIT <arm>_mixrN: PASS`; `tracking_metrics` prints a verdict + r0
10	"R sweep" (Tier C)	the `build_interleaved_run … --runs 1 --events 120 --total-ticks 26600 --run-offset 3` line from the runbook, then `rig_runcard --phase rsweep --run 4 …` → `build_r_schedule`, A (has `--r-schedule`), B, `run_audit`, `analyze_r_sweep`	`R SCHEDULE OK`; `AUDIT mpc_rsweep_r4: PASS`; per-block table with `rW0 ok` in every row
11	"stim zero" (end of day, or after ANY crash)	`.\rig\stim_zero.ps1`	`stim zero sent (8 words, 2 s of zeros)`
12	"archive"	`.\rig\archive_day.ps1 -Day 2026-10-22`; then `& $py rig\array_health.py --newest --context tissue --label "end of day"`	`DONE: …zip (N MB)` — then copy zip + blocks to the external drive; the array line goes in the notebook

Red flags — say them immediately:

- `ARRAY HEALTH: BROKEN` (any row) or the gate's `array:` line says BROKEN → the readout cannot be trusted; nothing after it means anything until the connector is reseated or the array swapped.
- the gate's `ADJUST:` says "move the REFERENCE" → the common-mode wave is flagged; the surgeon moves the reference, not the array.
- an IZ2 LED turns RED during a stim run → say "red LED, bank N" (compliance = open lead).
- the liveness window BEEPS or says `DEAD` (or `run_audit` says `y=DEAD`) → amplifier/PZ2 is off. Stop the run.
- `AUDIT …: FAIL` → read every `FAIL:` line aloud; B will refuse the next run until someone types `override`.
- any crash/abort of a stim run → `.\rig\stim_zero.ps1` BEFORE anything else.
- B never prints `policy=fresh` by tick ~100 → Ctrl+C in B, restart B, `go` again.
- `Skipping ... to position` in B → Synapse recording was started too early; stop, restart B, start recording LAST.
- A gate/ingest line prints `newest block:` with a time that is NOT the recording you just stopped → the copy has not landed; say so, it graded a STALE block.
- Any `FAIL` from `check_impulse_delivery` → read the `missed` count and time window aloud (a late stim-enable looks like missed probes at the start; that is operator timing, not a fault).
- The physio timer beeped (every 15 min) → read heart rate + body temperature off the PhysioSuite to the surgeon, type them (e.g. `HR 310 T 37.2`, or `-`) so the log keeps time.